

# Analysis plan (pre-registration)

## Weather, Monitors, or Policy? A meteorologically normalised, quasi-experimental evaluation of India's National Clean Air Programme, 2015–2025

Pre-registration, version of 2026-09-27. Source: docs/analysis\_plan.md at commit

6e24ecaf38c54c1f31774c966c243e4183c1b2ca of <https://github.com/breenu/ncap-evaluation>. Every number in this plan was produced by the project's pipeline from data up to 2018 only. References of the form DEC-*nnn* point to docs/DECISIONS.md, and file paths to the repository, at that commit.

## 1. Study information

**Title.** As above.

**Authors.** Reenu (sole author).

**Description.** NCAP, launched in January 2019, enrolled 131 Indian cities with a target of up to 40% lower PM10 by 2025–26. Public progress reports compare raw concentrations with a baseline. That comparison mixes three things: weather, changes in which monitors exist, and policy. This study separates them:

- **RQ1, measurement:** how much of a city's reported change comes from data-quality artefacts and network composition.
- **RQ2, weather:** how much comes from meteorology.
- **RQ3, policy:** the effect of NCAP enrolment on PM2.5 (and PM10), relative to comparable non-enrolled urban centres. This is the confirmatory question.
- **RQ4, heterogeneity and mechanism:** mostly exploratory.

**The central data limitation, stated first.** The ground network before NCAP is very small. Under the primary completeness rule, PM10 has 9 valid station-years in 2017 and 56 + 10 (NCAP + non-NCAP urban centres) in 2018. Non-NCAP centres have 0 valid PM10 station-years before 2018. So the ground layer has **one pre-NCAP year**, which cannot test parallel trends. Consequences:

- The **primary evidence is satellite PM2.5 (Layer A)**, which has nine pre-years (2010–2018).
- **Ground-based effects (Layer B), all PM10 effects, and the PM10-vs-PM2.5 mechanism test are secondary.** They carry that limitation beside every estimate.
- No minimum detectable effect is claimed for ground PM10 (§3).

### Hypotheses.

| #  | Status                                       | Hypothesis                                                                                                                                                                      | Layer |
|----|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| H1 | Confirmatory, primary                        | NCAP enrolment reduced annual population-weighted PM2.5 in enrolled urban centres, relative to comparable non-enrolled centres, 2019–2024 (2020 excluded).                      | A     |
| H2 | Confirmatory, tested only if H1 is supported | The reduction is larger in winter (Oct–Feb) than in the rest of the year. Most city measures and GRAP (the Graded Response Action Plan) target winter episodes.                 | A     |
| H3 | Secondary (mechanism)                        | If dust control dominated action, PM10 fell more than PM2.5 in enrolled cities, and the PM2.5/PM10 ratio rose.                                                                  | B     |
| H4 | Secondary (RQ1/RQ2; proposal hypothesis 1)   | Reported (raw, all-station) improvements in NCAP cities exceed deweathered, composition-corrected improvements: weather and network change explain part of the reported change. | B     |
| H5 | Secondary (RQ4; proposal hypothesis 3)       | Effects are smaller (less negative) in Indo-Gangetic Plain (IGP) cities, where regional sources dominate.                                                                       | A     |

H4 and H5 are hypotheses 1 and 3 of the original proposal (dated 25 September 2026, written before any data were downloaded). I left them out of my first draft and have restored them (see §6).

## 2. Design plan

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**Study type.** Observational; quasi-experimental; staggered adoption; secondary data only.

**Blinding.** I am not blinded in the usual sense: I am the only analyst and the data are public. The substitute is a **data-access rule** that I enforce in code (`config/gate.yaml`; `src/common/gate.py`):

- **What I had seen before writing this plan:**
  - all pre-2019 data;
  - post-2019 series for individual cities, regions and the whole network, descriptively;
  - data-quality diagnostics for every year.
- **My pre-gate computations** (all on data up to 2018; the code's data reader filters and asserts it):
  - the minimum detectable effect;
  - baseline balance;
  - a placebo on the real NCAP units at fake adoption years 2014 and 2015.
- **What I have not computed or plotted:** any comparison of NCAP with non-NCAP units after 2018.

**Study design: units, treatment, controls.**

- **Units (Layer A):** GHSL UCDB R2024A urban centres, one satellite unit per centre (DEC-006).
  - NCAP cities sharing a centre are one unit, e.g. Delhi, Faridabad, Ghaziabad and Noida.
  - **Asansol & Raniganj is one unit:** the Asansol centre joined with the GHSL centre UCDB names "Mejia", which holds Raniganj's built-up area (DEC-064, confirmed DEC-081).
- **Treated:** 113 units holding 121 NCAP cities.
- **Towns without a GHSL centre:** the 10 NCAP towns with none are **excluded from the primary estimate**, because treated and control units must be defined the same way (DEC-063). They enter one sensitivity analysis as 1.87 km buffered GeoNames points.
- **Patancheru (DEC-049): intention to treat.** It is enrolled from its listing date even though it is absent from the 2026 list, because dropping it would condition on a post-treatment event.
  - Layer A: its point lies inside the Hyderabad centre, which is treated anyway. The rule matters in the buffered-towns sensitivity analysis and on the ground (one station).
  - Sensitivity: exclude it.
- **Treatment timing, primary: first appearance on an official NCAP list.** Listed on or before 30 June counts as treated from that calendar year, otherwise from the next. Nothing is treated before 2019. A shared unit takes its earliest member's date.
  - Unit cohorts: 2019: 89; 2020: 15; 2021: 9.
- **Treatment timing, sensitivity only: the first financial year with a recorded central release** (my decision of 2026-09-27: XV-FC grants are partly performance-linked, so the timing of money is partly an outcome of air quality and must not define treatment), counted from the next calendar year (FY2019-20 → 2020; DEC-086). Sources: Lok Sabha AU2467 per-FY releases; XV-FC grants start in FY2020-21.
  - Unit cohorts: 2020: 86; 2021: 25; 2022: 2.
- **Anticipation:** 94 launch cities were already on CPCB's 2017 non-attainment list. Sensitivity: treat them from 2018.
- **Control pool:** non-NCAP centres with 2015 population  $\geq 100,000$ . 2015 is a pre-treatment epoch, so the pool cannot depend on post-treatment growth.

- Any centre containing the point of an NCAP town that has no centre of its own is excluded (Kalka, which contains Parwanoo).
- This leaves **923 controls**, from 1,810 non-NCAP centres (924 of them  $\geq 100k$ ).
- **Spillover:** primary = no buffer. Sensitivity = drop controls within 25 km, edge to edge, of any NCAP place, including the buffered towns. This leaves 754 controls (DEC-085).

| Rule                                                   | Units |
|--------------------------------------------------------|-------|
| GHSL urban centres in India (primary units)            | 1,923 |
| ... of which hold an NCAP city (treated units)         | 113   |
| Non-NCAP centres                                       | 1,810 |
| ... with 2015 population $\geq 100,000$                | 924   |
| ... not containing an NCAP town's point (control pool) | 923   |
| ... and at least 10 km from every NCAP place           | 852   |
| ... and at least 25 km from every NCAP place           | 754   |
| ... and at least 50 km from every NCAP place           | 511   |

- **Size of treated units:** they are kept whatever their size. Sensitivity: restrict them to  $\geq 100k$  too, which drops 4.

**Randomization.** None; treatment was assigned by policy. Enrolled cities were chosen *because* they were polluted, so selection and regression to the mean are the main threats. SDID's unit and time weights, the pre-trend tests and the placebos address them.

### 3. Sampling plan

**Existing data.** *Registration following analysis of the data.* I have downloaded, audited and described the data. I have not run the analyses that bear on the hypotheses (post-2018 NCAP vs non-NCAP comparisons), and a code-level gate prevents them from running.

**Explanation of existing data.** I checked every source and recorded it in manifests with checksums (docs/data-cards/):

- **Satellite:** ACAG V5.GL.06, V6.GL.03 and V6.GL.02.04.
- **Ground:** the CPCB mirror 2015–2025, plus OpenAQ for January–March 2026.
- **Weather:** ERA5.
- **Boundaries and population:** GHSL.
- **Treatment:** NCAP lists and funding tables from PDFs, with source page for every row.

Pre-period results I have already seen (details in [pregate\\_checks.md](#) §5): the real NCAP units show a placebo ATT of +0.47% at fake adoption 2014 (95% CI -0.24% to +1.19%; equal-tailed permutation  $p = 0.32$ ) and -0.41% at 2015 (95% CI -1.13% to +0.32%;  $p = 0.75$ ). Both are consistent with no pre-NCAP divergence.

**Data collection procedures.** None; all data are secondary. The pipeline is `snakemake pregate / all`.

**Sample size.**

- **Layer A:** 113 treated and 923 control units. Years 2010–2024: nine pre-years; the primary post-period is 2019 and 2021–2024.
- **Layer B:** NCAP-city stations with a valid 2018 baseline. In 2018 that is 56 PM10 station-years in 25 NCAP centres.

**Sample size rationale: minimum detectable effect (MDE).** The sample is fixed by the data, so I report what the design can detect.

- **Method:** placebo-in-time on 2010–2018 satellite data. 500 sets of 113 control units were given fake adoption in 2014 and in 2015, and SDID was re-estimated each time.
- **Two null designs:** random sets, and region-matched sets (the same regional make-up as the real treated set).
- **Formula:**  $SE = SD \text{ of the placebo ATTs}$ ;  $MDE = 2.8 \times SE$  (5% two-sided test, 80% power). The largest over both designs and both fake years is used.

| Outcome                                          | Null design    | Fake adoption | SE                            | Mean placebo ATT               | MDE                           |
|--------------------------------------------------|----------------|---------------|-------------------------------|--------------------------------|-------------------------------|
| Annual PM2.5, log (primary)                      | random         | 2014          | 0.0042                        | +0.0001                        | 0.0117 (1.2% fall)            |
| Annual PM2.5, log (primary)                      | region-matched | 2014          | 0.0036                        | +0.0011                        | 0.0102 (1.0% fall)            |
| Annual PM2.5, log (primary)                      | random         | 2015          | 0.0041                        | +0.0002                        | 0.0115 (1.1% fall)            |
| Annual PM2.5, log (primary)                      | region-matched | 2015          | 0.0037                        | -0.0028                        | 0.0104 (1.0% fall)            |
| Annual PM2.5, level ( $\mu\text{g}/\text{m}^3$ ) | random         | 2014          | $0.27 \mu\text{g}/\text{m}^3$ | $+0.00 \mu\text{g}/\text{m}^3$ | $0.75 \mu\text{g}/\text{m}^3$ |
| Annual PM2.5, level ( $\mu\text{g}/\text{m}^3$ ) | region-matched | 2014          | $0.24 \mu\text{g}/\text{m}^3$ | $-0.36 \mu\text{g}/\text{m}^3$ | $0.66 \mu\text{g}/\text{m}^3$ |
| Annual PM2.5, level ( $\mu\text{g}/\text{m}^3$ ) | random         | 2015          | $0.33 \mu\text{g}/\text{m}^3$ | $+0.00 \mu\text{g}/\text{m}^3$ | $0.93 \mu\text{g}/\text{m}^3$ |
| Annual PM2.5, level ( $\mu\text{g}/\text{m}^3$ ) | region-matched | 2015          | $0.29 \mu\text{g}/\text{m}^3$ | $-0.48 \mu\text{g}/\text{m}^3$ | $0.81 \mu\text{g}/\text{m}^3$ |
| Winter PM2.5 (Oct-Feb), log                      | random         | 2014          | 0.0059                        | +0.0001                        | 0.0165 (1.6% fall)            |
| Winter PM2.5 (Oct-Feb), log                      | region-matched | 2014          | 0.0057                        | -0.0020                        | 0.0158 (1.6% fall)            |
| Winter PM2.5 (Oct-Feb), log                      | random         | 2015          | 0.0055                        | +0.0001                        | 0.0155 (1.5% fall)            |
| Winter PM2.5 (Oct-Feb), log                      | region-matched | 2015          | 0.0053                        | -0.0051                        | 0.0148 (1.5% fall)            |
| Non-winter PM2.5 (Mar-Sep), log                  | random         | 2014          | 0.0054                        | +0.0001                        | 0.0151 (1.5% fall)            |
| Non-winter PM2.5 (Mar-Sep), log                  | region-matched | 2014          | 0.0053                        | +0.0003                        | 0.0147 (1.5% fall)            |
| Non-winter PM2.5 (Mar-Sep), log                  | random         | 2015          | 0.0059                        | +0.0002                        | 0.0164 (1.6% fall)            |
| Non-winter PM2.5 (Mar-Sep), log                  | region-matched | 2015          | 0.0055                        | -0.0048                        | 0.0155 (1.5% fall)            |
| Winter minus non-winter, log (H2)                | random         | 2014          | 0.0072                        | -0.0000                        | 0.0201 ( $\approx 2.0$ pp)    |
| Winter minus non-winter, log (H2)                | region-matched | 2014          | 0.0077                        | -0.0023                        | 0.0215 ( $\approx 2.1$ pp)    |
| Winter minus non-winter, log (H2)                | random         | 2015          | 0.0065                        | -0.0001                        | 0.0182 ( $\approx 1.8$ pp)    |
| Winter minus non-winter, log (H2)                | region-matched | 2015          | 0.0068                        | -0.0003                        | 0.0191 ( $\approx 1.9$ pp)    |

**The MDE for H1 is a 1.2% fall in annual population-weighted PM2.5** (0.0117 in natural-log units). At the NCAP units' 2010–2018 mean of  $51.7 \mu\text{g}/\text{m}^3$  that is about  $0.6 \mu\text{g}/\text{m}^3$ . A separate fit on the  $\mu\text{g}/\text{m}^3$  scale gives  $0.9 \mu\text{g}/\text{m}^3$ : absolute noise is concentrated in the most polluted units, so the  $\mu\text{g}/\text{m}^3$  MDE is not the % MDE times the mean (pregate\_checks.md §4). The primary outcome is on the log scale, so the % MDE is the one that counts. The MDE for H2 (winter minus non-winter) is 0.0215 in log units ( $\approx 2.1$  percentage points). **Read this MDE as a lower bound, not a power calculation.** ACAG PM2.5 is a smooth, calibrated product averaged over a hundred units, so pre-period noise is small. The real design is noisier:

- its post-years reach 6 years after adoption, where synthetic controls drift more (the placebo post-periods are 4–5 years);
- the placebo assigns every unit at once, while the real estimator averages cohorts;
- even region-matched sets are more dispersed than the real treated set;
- the post-period holds shocks the pre-period does not (COVID, BS-VI).

The confidence intervals use the real design's own SE (joint placebo, §5). I report the MDE for information only: it is a best-case noise level, not the smallest effect that matters, so it is **not** the equivalence margin. The margin

is a substantive smallest effect of interest,  $\pm 5\%$  (\$5).

**Ground PM10: no MDE.** Non-NCAP centres have no valid PM10 before 2018, so no pre-period change exists on the control side. Only 6 stations in 5 NCAP cities are valid in both 2017 and 2018. A placebo is therefore impossible, and a variance from one year-pair would not be an MDE.

**Stopping rule.** None (fixed existing data).

## 4. Variables

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**Manipulated variables.** None. The treatment indicator is NCAP enrolment (timing as in §2).

**Measured variables.**

- **Layer A outcome (primary):** log annual population-weighted mean PM2.5 over the unit: ACAG V5.GL.06, 0.01°, weighted by GHS-POP 2020 (DEC-070, DEC-081). Sensitivity: the area-weighted mean.
- **Seasonal outcomes (H2):** V5.GL.06 monthly.
  - Winter season-year  $t$  = October  $t$  to February  $t+1$ , so winter 2019 is the first fully after launch and winter 2023 the last complete one.
  - Non-winter  $t$  = March to September  $t$ .
  - A season needs all its months.
- **Layer B outcome:** station daily PM2.5 and PM10 → deweathered (Phase 5) → city-year on a balanced station panel (Phase 6).
  - Balanced-panel baseline 2018 (primary), 2019 (sensitivity).
  - Ground years: **calendar 2025 is the last primary year.** January–March 2026 exists only in OpenAQ's raw real-time feed, not CPCB's validated archive (DEC-079), so it enters a FY2025-26 sensitivity analysis only.
- **Ground validity rules:**
  - Valid hour: at least 1 of 4 quarter-hours (primary); at least 3 of 4 (sensitivity) (DEC-069).
  - Valid day and valid year: 75% of hours and 75% of days (primary); 60% and 90% (sensitivity).
  - Flagged values are removed before daily means: ceilings, flatlines of 4 or more hours, PM2.5 > PM10 (DEC-068).
- **Covariates:**
  - ERA5 monthly meteorology averaged over each unit: temperature, humidity, wind, boundary-layer height, precipitation, radiation.
  - Region × year effects, with regions IGP, coastal, north-east and peninsular/other (DEC-075).
  - VIIRS fire radiative power, from 2012, in one sensitivity check (DEC-039).

**Indices.**

- The station reliability score (0–100; DEC-073) is descriptive. It is used only in the "exclude score < 50" sensitivity analysis.
- Deweathered series: the model family with the better median out-of-sample  $R^2$  under blocked, forward-chaining CV is primary; the other is the sensitivity (DEC-088). Choosing on CV fit is blind to treatment.

## 5. Analysis plan

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**Statistical models.**

1. **Primary: synthetic difference-in-differences** (SDID; Arkhangelsky et al. 2021; R synthdid).

- Estimated separately per listing cohort with never-treated controls, since the package needs simultaneous adoption (DEC-027).
- Pre-period: 2010 to the year before the cohort's adoption. Post-period: adoption to 2024.
- **The year 2020 is dropped from the primary panel.**
- Cohort ATTs are combined weighted by their number of treated units.
- **SE by joint placebo:** in each of 500 replications, disjoint random sets of controls, of the cohorts' sizes, are given the cohorts' adoption years; the aggregate is re-estimated each time. 95% CI =  $ATT \pm 1.96 SE$ .

## 2. Event study.

- Unit effects, region  $\times$  year effects, ERA5 covariates, and relative-year indicators from  $-9$  to  $+5$  with  $-1$  omitted.
- Estimated with the Sun & Abraham (2021) interaction-weighted estimator against never-treated units, which avoids the bias of two-way fixed effects under staggered adoption.
- SEs clustered by unit. The 2020 coefficient is shown separately.
- **Sensitivity to pre-trends (reported, not a decision rule):** Rambachan & Roth (2023) bounds from R HonestDiD (0.2.8, in the same 2026-09-25 CRAN snapshot; installed in Phase 7), applied to the event-study coefficients and their clustered covariance. I report two restrictions for the average post-period effect: relative magnitudes (post-period violations at most  $\bar{M}$  times the largest pre-period one;  $\bar{M} = 0, 0.5, 1, 1.5, 2$ ) and smoothness (a limit  $M$  on how much the trend's slope may change between consecutive periods;  $M$  from 0 to twice the largest pre-period coefficient's SE, in 5 equal steps). I report the robust 95% CIs and the breakdown value: the largest  $\bar{M}$  at which the CI still excludes 0.

## 3. Callaway & Sant'Anna (2021), R did: doubly robust, never-treated controls (primary for this estimator) and not-yet-treated (sensitivity). Simple and dynamic aggregations, with uniform bands.

## 4. Layer B.

- **ITS:** deweathered interrupted time series per NCAP city with valid 2018 data. It is a before–after contrast of deweathered series and cannot separate national shocks.
- **Ground DiD:** NCAP vs non-NCAP stations with year effects, only where both have 2018 data.
- Both are reported for PM2.5 and PM10 separately; 95% CIs by cluster bootstrap over cities (1,000 resamples).

## 5. Hierarchical model (Phase 8): a measurement-error model of city effects and their SEs, with moderators: baseline PM2.5, IGP, log population, coastal, funding channel. This gives shrunken city estimates. H5 is the IGP coefficient.

**Transformations.** Natural log of concentrations; effects reported as % change ( $100 \times (e^{\beta} - 1)$ ), with  $\mu\text{g}/\text{m}^3$  from the level-scale fit as secondary.

**Inference criteria and decision rules** (unchanged from my first draft except where §6 says otherwise).

- **H1 supported** if all of these hold:
  - (a) the primary SDID ATT is negative and its 95% CI excludes 0;
  - (b) the event-study pre-period coefficients ( $-9$  to  $-2$ ) are jointly insignificant (Wald  $p > 0.10$ );
  - (c) the 2016 placebo-in-time ATT's 95% CI includes 0;
  - (d) the estimate keeps its sign in Callaway & Sant'Anna, in the area-weighted outcome and in V6.GL.03.
- **If (b) or (c) fails:** "not identified by this design", whatever the ATT. If (b) fails, I report the HonestDiD bounds alongside, showing how large a pre-trend violation the estimate could withstand; they do not overturn the verdict.

- **If the CI includes 0:** "no detectable effect", never "no effect". Plus an equivalence test (two one-sided tests at 5%) against a **smallest effect of interest of  $\pm 5\%$  in annual PM<sub>2.5</sub>**: if the 90% CI lies within  $\pm 5\%$  (in log units, between  $\ln 0.95$  and  $\ln 1.05$ ), effects of that size or larger are ruled out; otherwise the result is inconclusive. Why  $\pm 5\%$ : one quarter of NCAP's smallest target (a 20% reduction). That target was set for PM<sub>10</sub>, not PM<sub>2.5</sub>, so the margin is a judgement, not an official threshold. The MDE (1.2%) is still reported.
- **Whatever the outcome:** report which effect sizes the 95% CI excludes, set against NCAP's own targets (a 20–30% reduction, later up to 40%, in PM<sub>10</sub>). A satellite PM<sub>2.5</sub> result says nothing directly about PM<sub>10</sub> attainment, and I will say so.
- **H2** (tested only if H1 is supported): the winter-minus-non-winter ATT difference, from the same joint placebo draws. Supported if the 95% CI excludes 0 in the direction of a larger winter reduction. If H1 is not supported, I report H2 as exploratory.
- **H3:** labelled "consistent with dust control" only if all of these hold:
  - the PM<sub>2.5</sub>/PM<sub>10</sub> ratio rises, with a 95% CI excluding 0;
  - the PM<sub>10</sub> effect exceeds the PM<sub>2.5</sub> effect in  $\geq 2$  of 3 specifications (raw, deweathered, balanced panel);
  - Layer A is not in the opposite direction.

Otherwise "inconclusive". Never "confirmed", because one pre-year cannot test parallel trends.

- **H4:** per NCAP city with ground data, raw all-station change minus deweathered balanced-panel change, from 2018 to 2025. Supported if the mean across cities is positive with a 95% cluster-bootstrap CI excluding 0. This is descriptive (NCAP cities only), not causal.
- **H5:** supported if the IGP coefficient's 95% credible interval lies above 0 (a less negative effect).
- **Robustness:** a check "agrees" if it has the primary's sign and its point estimate lies inside the primary 95% CI. I report every check in one table, whichever way it points.

**Threat: satellite calibration leakage** (pre-specified sensitivity check). ACAG calibrates satellite PM<sub>2.5</sub> to ground monitors, and NCAP added monitors mainly in treated cities, so a Layer A effect could partly reflect calibration rather than air quality.

- **Split.** I estimate the primary SDID (same controls, cohorts, 2020 handling and joint-placebo SE) separately for treated units that gained a CAAQMS station inside their polygon with first PM data in 2019–2024 (the satellite post-period) and for those that did not. The difference (gained minus not gained) gets its SE from joint placebo draws of both group sizes.
- **Group sizes** (network metadata only, `pregate_checks.md` §7): 74 units gained a monitor; 39 did not (12 already had a station before 2019, 27 never had one inside the polygon). A group with fewer than 10 units is not estimated, and I say that it is too small; with these sizes, both groups are large enough to estimate.
- **Warning sign of leakage**, which I report as such beside H1: the gained group's ATT is negative with a 95% CI excluding 0 while the not-gained group's CI includes 0, or the difference is negative with a 95% CI excluding 0. It is a warning, not proof: cities that gained monitors also differ in size and pollution, so the gap could be real heterogeneity.
- **Related evidence under the same threat:** the V6.GL.02.04 vintage comparison (calibrated to an earlier monitor set), and, if time allows, raw MAIAC aerosol optical depth, which uses no ground monitors.

## Multiple comparisons.

- **Confirmatory family = {H1, H2}, tested in fixed sequence at  $\alpha = 0.05$ .** H2 is tested only if H1 is supported. This keeps the familywise error rate at 5% without splitting  $\alpha$ .
- **H3–H5 are secondary:** each at 0.05, labelled secondary, never promoted to confirmatory.

- **Robustness checks and event-study coefficients** are not hypothesis tests. The event study gets one joint pre-trend test, plus pointwise CIs and uniform (simultaneous) bands from the CS estimator.
- **City-level claims** come from the hierarchical model's shrunken estimates. Any unshrunk city claim is held to Benjamini–Hochberg FDR 5% across all 113 treated units. There is no unshrunk "best/worst cities" table.
- **Moderators other than IGP** are exploratory.

### Triangulation, and how disagreement between layers is reported.

- Layer A remains the headline. I show Layer B beside it and never average the two.
- The like-for-like comparison is Layer A restricted to the units that have Layer B stations, PM2.5 only.
- The pair is classified in advance as one of:
  - **consistent**: same sign and overlapping 95% CIs;
  - **different magnitude**: same sign, non-overlapping CIs;
  - **conflict**: opposite signs with at least one CI excluding 0;
  - **uninformative**: the Layer B CI contains both 0 and the Layer A estimate.
- **Any category other than "consistent" triggers a fixed investigation**, and I report each step whether or not it closes the gap:
  1. network composition (all stations vs balanced panel);
  2. satellite calibration to the ground (V6.GL.02.04 vintage vs V5.GL.06; the ground–satellite correlation over time);
  3. spatial coverage (station sites vs the population-weighted polygon);
  4. deweathering (raw vs deweathered).
- I never treat ground PM10 and satellite PM2.5 as "agreement".

### Primary choices and their sensitivity checks (I run and report every one).

| Choice                           | Primary                           | Sensitivity                                                                                                | Log              |
|----------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------|------------------|
| Satellite product                | V5.GL.06                          | V6.GL.03 (algorithm); V6.GL.02.04 (vintage, to 2023)                                                       | DEC-001/002      |
| Unit value                       | population-weighted mean          | area-weighted mean                                                                                         | DEC-070/081      |
| Towns without a GHSL centre (10) | excluded                          | included as 1.87 km buffers                                                                                | DEC-063          |
| Patancheruvu                     | intention to treat                | excluded                                                                                                   | DEC-049          |
| Asansol & Raniganj               | one unit incl. the "Mejia" centre | Asansol centre alone (added 2026-09-27)                                                                    | DEC-064/081      |
| Treated-unit size                | all                               | ≥ 100k only                                                                                                | first draft      |
| Spillover                        | none                              | 25 km buffer                                                                                               | DEC-085          |
| Treatment date                   | first listing                     | first funding year (never primary: funding is partly performance-linked); 2018 for the 94 on the 2017 list | DEC-086, DEC-092 |
| 2020                             | dropped                           | included; own coefficient                                                                                  | first draft      |
| Controls (CS)                    | never-treated                     | not-yet-treated                                                                                            | —                |
| Valid hour (ground)              | ≥ 1 of 4 quarter-hours            | ≥ 3 of 4                                                                                                   | DEC-069          |
| Completeness (ground)            | 75% / 75%                         | 60%, 90%                                                                                                   | DEC-073          |
| Ground end                       | calendar 2025                     | + Jan–Mar 2026 (FY2025-26, provisional)                                                                    | DEC-079          |

| Choice                                     | Primary                 | Sensitivity                                                                                                                                              | Log      |
|--------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Station coordinates I set by judgement (5) | recommended coordinates | drop all 5                                                                                                                                               | DEC-080  |
| Balanced-panel baseline                    | 2018                    | 2019                                                                                                                                                     | DEC-088  |
| Deweathering                               | better-CV family        | other family; none                                                                                                                                       | DEC-088  |
| Reliability                                | all valid station-years | drop score < 50                                                                                                                                          | DEC-073  |
| Calibration leakage                        | all treated units       | split by gained a monitor 2019–2024 or not; MAIAC AOD if time allows                                                                                     | DEC-096  |
| Pre-trends                                 | rule (b)                | HonestDiD bounds (relative magnitudes, smoothness)                                                                                                       | DEC-097  |
| Other checks                               | —                       | placebo in time (2016); placebo in space (500 permutations); leave-one-out donors; exclude IGP; Himalayan region split; VIIRS fire covariate (from 2012) | proposal |

**Data exclusion.** Only by the rules above (flags, completeness, control-pool rules). No outlier exclusion on outcome values.

#### Missing data.

- The satellite panel is complete (no missing unit-years).
- The completeness rules handle ground gaps; I do not impute in the primary analysis.
- The audit found missingness is *lower* on polluted days, which biases naive annual means slightly upward (median +0.7%, `audit_report.md`). I report this beside Layer B.

#### Exploratory analyses (I label them as such):

- funding dose-response on *allocations*, with the reverse-causality caveat (cut item 1);
- moderators other than IGP;
- NO<sub>2</sub> as a secondary pollutant.

(Raw MAIAC AOD is listed under the calibration-leakage threat above, still conditional on time.)

## 6. Other

#### Changes I made to my first draft after computing the pre-gate numbers (listed here for transparency):

1. Numbers filled in: MDE, pool, balance, cohorts.
2. H4 and H5 restored. They are hypotheses 1 and 3 of the original proposal dated 25 September 2026, written before any data were downloaded (the proposal PDF was created 2026-09-25 18:26 UTC and committed in the repository's first commit; the earliest raw download in any manifest is 2026-09-26 04:44 UTC). So they were not chosen after seeing data.
3. H2 made conditional on H1 (fixed sequence).
4. The joint-placebo SE for the cohort aggregate; never-treated donors in each per-cohort SDID.
5. The season-year definition.
6. The Sun & Abraham event-study estimator.
7. The layer-disagreement categories.
8. The deweathering-family rule.
9. The funding-date rule.
10. The Asansol-alone sensitivity (added 2026-09-27).

11. Units and p-values corrected (DEC-091): I give log-scale results in natural-log units with the implied percentage (my draft's "log points" label could be read as 100 times too large), permutation p-values are equal-tailed, and the % and  $\mu\text{g}/\text{m}^3$  MDEs are reported separately. No estimate changed.

**Further changes I made on 2026-09-27, before OSF registration:**

1. **Equivalence margin:** a smallest effect of interest of  $\pm 5\%$  in annual  $\text{PM}_{2.5}$  (one quarter of NCAP's smallest target, 20%, which was set for  $\text{PM}_{10}$ ) replaces  $\pm \text{MDE}$ . The MDE is a best-case noise level, so with it as the margin a null would almost always be "inconclusive". The MDE is still reported (DEC-095).
2. **Calibration-leakage check:** H1 split by whether treated units gained a CAAQMS station in the satellite post-period, with group sizes, a minimum group size and a pre-stated warning rule; MAIAC AOD listed under this threat, conditional on time (DEC-096).
3. **HonestDiD bounds** on the event study as a reported sensitivity analysis; rule (b) unchanged, and if (b) fails the bounds are reported alongside "not identified" (DEC-097).

H1's decision rules (a)–(d) are unchanged.

**Baseline balance** (unweighted, 2010–2018). NCAP units are much larger than the pool (log-population SMD 1.44) and less often in the IGP (28% vs 43%). Their mean  $\text{PM}_{2.5}$  is similar (51.7 vs  $56.4 \mu\text{g}/\text{m}^3$ ), and so is their pre-trend (2.3 vs 2.1% a year). SDID does not need level balance, only a matched pre-period path, which the placebos test.

| Characteristic (2010-2018 unless stated)                                     | NCAP units (n=113) | Control pool (n=923) | SMD   | Buffered pool (n=754) | SMD (buffered) |
|------------------------------------------------------------------------------|--------------------|----------------------|-------|-----------------------|----------------|
| $\text{PM}_{2.5}$ mean 2010-2018 ( $\mu\text{g}/\text{m}^3$ )                | 51.67              | 56.37                | -0.22 | 54.30                 | -0.12          |
| $\text{PM}_{2.5}$ trend 2010-2018 (% per year)                               | 2.28               | 2.11                 | 0.17  | 2.11                  | 0.18           |
| Winter $\text{PM}_{2.5}$ mean 2010-2017 seasons ( $\mu\text{g}/\text{m}^3$ ) | 72.09              | 80.80                | -0.23 | 76.98                 | -0.13          |
| Population 2015 (log10)                                                      | 5.94               | 5.32                 | 1.44  | 5.31                  | 1.46           |
| Area (km <sup>2</sup> )                                                      | 211.58             | 48.26                | 0.53  | 43.81                 | 0.55           |
| Region: coastal                                                              | 15.9%              | 13.9%                | 0.06  | 14.7%                 | 0.03           |
| Region: igp                                                                  | 28.3%              | 43.2%                | -0.31 | 39.5%                 | -0.24          |
| Region: north-east                                                           | 5.3%               | 4.4%                 | 0.04  | 4.5%                  | 0.04           |
| Region: peninsular/other                                                     | 50.4%              | 38.5%                | 0.24  | 41.2%                 | 0.19           |

**Deviations.** I will log any deviation from this plan in docs/DECISIONS.md, dated and justified, and list it in the final report.